

DEVI AHILYA UNIVERSITY

INSTITUTE OF ENGINEERING AND TECHNOLOGY, INDORE

SUBJECT CODE – 3VLRC3; EXAM CODE – 2024047 SUBJECT NAME – SURVEYING

DATE – Dec 2023; BE II YR. CIVIL ENGG.

TIME – 3 HRS

Note: Each question carries 6 marks. In Q.3 all parts carry 4 marks each.

Q.1) (a) What do you mean by Surveying? Explain principles of Surveying?

Q.1) (b) Explain difference types of chain used in surveying?

OR

Q.1) (c) Explain in Brief various Classification of Surveying with its uses.

Q.1) (d) What do you mean by Ranging Rod, Tape, Arrow Explain in Brief.

Q.2) (a) The following consecutive readings were taken with a dumpy level along a chain line at a common interval of 15m. The first reading was at a chainage of 165m where the RL is 98.085. The instrument was shifted after the 3rd & 9th readings.

3.150, 2.245, 1.125, 0.860, 3.125, 2.760, 1.835, 1.470, 1.965, 1.225, 2.390 & 3.035m

Mark rules on a page of your notebook in the form of a level book page and enter on it the above readings and find the RL of all the points by Rise and Fall Method.

Q.2) (b) Define the following terms:

Fore Sight, Back Sight, Bench Mark, Reduced Level.

OR

Q.2) (c) The following consecutive readings were taken with a leveling instrument at intervals of 20m. The first reading was at a chainage of 200m where the RL is 110.200m.

2.375, 1.730, 0.615, 3.450, 2.835, 2.070, 1.835, 0.985, 0.435, 1.630, 2.255 and 3.630m

The instrument was shifted after the 4th & 8th readings. Find the RLs of all the points using Height of Instrument Method.

Q.2) (d) What are the different types of leveling, Explain in Brief.

Q.3) (a) What is the difference between prismatic & surveyor's compass?

Q.3) (b) Convert the following WCB to QB ?

(i) $22^{\circ}30'$ (ii) $170^{\circ}12'$ (iii) $211^{\circ}54'$ (iv) $327^{\circ}24'$

Q.3) (c) Convert the following QB to WCB?

(i) N $12^{\circ}24'$ E (ii) S $31^{\circ}36'$ E (iii) S $68^{\circ}6'$ W (iv) N $5^{\circ}42'$ W

OR

Q.3) (d) Explain Different types of Meridians with Sketch.

Q.3) (e) The Magnetic Bearing of a line AB is $S 50^{\circ}15' W$. Find its True bearing, if the declination is $15^{\circ}25' E$.

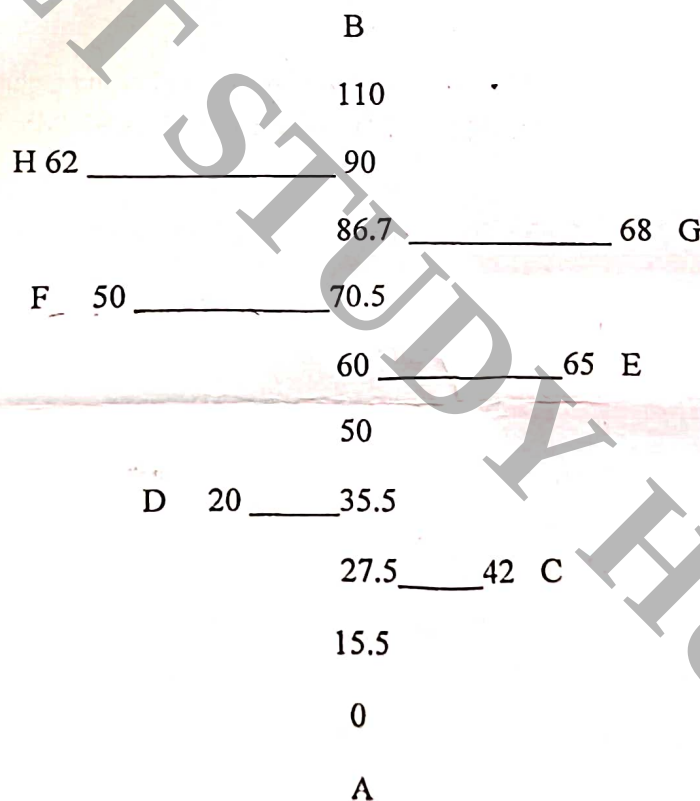
Q.3) (f) The Fore Bearings of the lines AB, BC, CD and DE are $55^{\circ}30'$, $125^{\circ}15'$, $230^{\circ}30'$, and $295^{\circ}45'$, respectively. Find angles B, C and D.

Q.4) (a) A series of offsets were taken from a chain line to an irregular boundary line at intervals of 15m in the following order. 0, 2.65, 3.80, 3.75, 4.65, 3.60, 4.95, 5.85 m. Compute the area between the chain line, the irregular boundary line and the end offsets by (a) Mid Ordinate Rule (b) Average ordinate rule (c) Trapezoidal rule (d) Simpson's rule

Q.4) (b) Write difference between Trapezoidal and Simpson's rule.

OR

Q.4) (c) A page of the field book of a cross staff survey is given in Fig. plot the required figure & calculate the relevant area.



Q.4) (d) Define Planimeter and procedure for finding out area from it.

Q.5) (a) The Ground level along the Centre line of a road is given below: 9213.7083

Chainage (m)	0	100	200	300	400	500	600
GL (m)	118.5	117.25	116.95	115.65	116.20	117.20	118.05

It is proposed that the formation level of RL 116.00 should be kept constant of starting from the chainage 0m. The formation width of the road is 13m and the side slope is 1:1, The ground is level transverse to the centerline. Calculate the volume of earthwork by prismoidal formula?

Q.5) (b) A railway embankment is 12 m wide with side slopes 1.5 to 1. Assuming the ground to be level in a direction transverse to the Centre line, calculate the volume contained in a length of 120 m, the

Centre heights at 30 m intervals being in meter are 1.90, 2.25, 3.15, 3.50, 2.85, 2.35, 1.85. Calculate the volume of earth work according to Both formulae.

OR

Q.5) (c) A railway embankment of formation width of 10m with side slopes 2 : 1 is to be constructed. The Ground level along the Centre line is given below:

Chainage (m)	0	50	100	150	200	250
GL (m)	116.75	115.35	117.85	116.20	119.50	119.25

The embankment has a rising gradient of 1 in 100, and the formation level at zero chainage is 115.50. Assuming the ground is level across the centerline, Compute the volume of the earth work?

Q.5) (d) The Area enclosed by the contour in a lake is given below:

Contour (m)	180	190	200	210	220	230	240
Area (m ²)	2,130	8,900	18,500	27,300	35,000	41,200	55,300

Calculate the volume of water between the contours 180m and 240m by: (1) Trapezoidal formula, (2) Prismoidal formula.

$$A = (b + sh)S$$